STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 16

PROBABILITY DISTRIBUTIONS

MULTINOMIAL DISTRIBUTION

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MULTINOMIAL DISTRIBUTION

Recall that in binomial distribution, each trial has two outcomes. But if each trial in an experiment has more than two outcomes, then we use **multinomial distribution.** *So the binomial experiment becomes a multinomial experiment if we let each trial have more than two possible outcomes.*

THE MULTINOMIAL EXPERIMENT A multinomial experiment is a probability experiment that satisfies the following four requirements

1- There must be a fixed number of trials. *i.e*. there are *n* identical trials.
2- Each trial has more than two possible outcomes.
3- The trials are independent of each other.
4- The probability of success for each outcome remains same for each trial.

EXAMPLES OF MULTINOMIAL EXPERIMENT Some examples of multinomial experiment are as follows

1. A survey might require the responses of "approve," "disapprove," or "no opinion."
2. Samples of manufactured products are rated excellent, above average, average, or inferior
3. A person may have a choice of one of five activities for Friday night, such as a movie, dinner, baseball game, play, or party.

THE MUTINOMIAL DISTRIBUTION Multinomial experiment and its results give rise to a special probability distribution called the multinomial distribution.

Consider a sequence of *n* independent trials where each individual trial can have *k* outcomes that occur with constant probability values $p_1, p_2, p_3,..., p_k$, then the probability distribution of the random variables $X_1, X_2, X_3,..., X_k$, representing number of occurrences for the events $E_1, E_2, E_3,..., E_k$ said to have a **multinomial distribution,** and their joint probability mass function is given by

$$P(X_1 = x_1, X_2 = x_2,..., X_k = x_k) = \frac{n!}{x_1!.x_2!...x_k!} \times {p_1}^{x_1} \times {p_2}^{x_1} \times \ldots {p_k}^{x_k}$$

where $x_1 + x_2 + x_3 + ... + x_k = n$ *i.e.* $\sum_{i=1}^{k} x_i = n$

and $p_1 + p_2 + p_3 + ... + p_k = 1$ *i.e.* $\sum_{i=1}^{k} p_i = 1$

PROPERTIES OF MULTINOMIAL DISTRIBUTION

1- The mean of Multinomial Distribution is $E(X_i) = n\,p_i$

2- The variance of Multinomial Distribution is $V(X_i) = n\,p_i\,q_i$

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EXAMPLES OF MUTINOMIAL DISTRIBUTION

EXAMPLE-1 A die is rolled 5 times. Find the probability that 1 appears twice, 6 appears once and any other number appears twice. Also find expectation and variance of getting 6.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote the random variable that* 1 *appear.*

Let X_2 *denote the random variable that* 6 *appear.*

Let X_3 *denote the random variable that any number other than* 1 *and* 6 *appear.*

Here $n=4,\ X_1=2,\ X_2=1,\ X_3=2,\ p_1=1/6,\ p_2=1/6,\ p_3=4/6$

Then $P(X_1=2,\ X_2=1,\ X_3=2)=\dfrac{5!}{2!\times1!\times2!}\left(\dfrac{1}{6}\right)^2\left(\dfrac{1}{6}\right)^1\left(\dfrac{4}{6}\right)^2=0.062$

Then $E(X_2)=np_1=(5)(1/6)=0.83,\ V(X_2)=np_2q_2=(5)(1/6)(5/6)=0.69$

EXAMPLE-2 In a large city, 50% of the people choose a movie, 30% choose dinner and a play, and 20% choose shopping as a leisure activity. If a sample of 5 people is randomly selected, find the probability that 3 are planning to go to a movie, 1 to a play, and 1 to a shopping mall.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote the people who choose movie*

Let X_2 *denote the people who choose dinner and play*

Let X_3 *denote the people who choose shopping*

Here $n=5,\ X_1=3,\ X_2=1,\ X_3=1,\ p_1=0.50,\ p_2=0.30,\ p_3=0.20$

Then $P(X_1=3,\ X_2=1,\ X_3=1)=\dfrac{5!}{3!\times1!\times1!}(0.50)^3(0.30)^1(0.20)^1=0.15$

EXAMPLE-3 A box contains 4 white balls, 3 red balls, and 3 blue balls. A ball is selected at random, and its color is written down. It is replaced each time. Find the probability that if 5 balls are selected, 2 are white, 2 are red, and 1 is blue.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote the number of white balls selected*

Let X_2 *denote the number of red balls selected*

Let X_3 *denote the number of blue balls selected*

Here $n=5,\ X_1=2,\ X_2=2,\ X_3=1,\ p_1=4/10,\ p_2=3/10,\ p_3=3/10$

Then $P(X_1=2,\ X_2=2,\ X_3=1)=\dfrac{5!}{2!\times2!\times1!}\left(\dfrac{4}{10}\right)^2\left(\dfrac{3}{10}\right)^2\left(\dfrac{3}{10}\right)^1=\dfrac{81}{625}$

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EXAMPLE-4 A certain city has 3 newspapers, A, B, and C. Newspaper A has 50 percent of the readers in that city. Newspaper B, has 30 percent of the readers, and newspaper C has the remaining 20 percent. Find the probability that, among 8 randomly-chosen readers in that city, 5 will read newspaper A, 2 will read newspaper B, and 1 will read newspaper C.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote readers who read newspaper A*

Let X_2 *denote readers who read newspaper B*

Let X_3 *denote readers who read newspaper C*

Here $n=8,\ X_1=5,\ X_2=2,\ X_3=1,\ p_1=0.5,\ p_2=0.3,\ p_3=0.2$

Then $P(X_1=5,\ X_2=2,\ X_3=1)=\dfrac{8!}{5!\times 2!\times 1!}(0.50)^5(0.30)^2(0.20)^1=0.0945$

EXAMPLE-5 If a pair of dice is rolled 6 times. What is the probability of obtaining a total of 7or 11 twice, a matching pair once and any other combination 3 times.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote the random* var*iable that a total of* 7 *or* 11 *is obtained*

Let X_2 *denote the random* var*iable that a matching pair is obtained*

Let X_3 *denote the random* var*iable that any other combination is obtained*

Here $n=6,\ X_1=2,\ X_2=1,\ X_3=3,\ p_1=\dfrac{2}{9},\ p_2=\dfrac{1}{6},\ p_3=\dfrac{11}{18}$

Then $P(X_1=2,\ X_2=1,\ X_3=3)=\dfrac{6!}{2!\times 1!\times 3!}\left(\dfrac{2}{9}\right)^2\left(\dfrac{1}{6}\right)^1\left(\dfrac{11}{18}\right)^3=0.1127$

EXAMPLE-6 A small airport coffee shop manager found that the probabilities a customer buys 0, 1, 2, or 3 cups of coffee are 0.3, 0.5, 0.15, and 0.05, respectively. If 8 customers enter the shop, find the probability that 2 will purchase something other than coffee, 4 will purchase 1 cup of coffee, 1 will purchase 2 cups, and 1 will purchase 3 cups.

SOLUTION *Let us define the RandomVariables first.*

Let X_1 *denote the customers who purchase no coffee.*

Let X_2 *denote the customers who purchase one cup of coffee.*

Let X_3 *denote the customers who purchase two cups of coffee*

Let X_4 *denote the customers who purchase three cups of coffee*

Here $n=8,\ X_1=2,\ X_2=4,\ X_3=1,\ X_4=1,\ p_1=0.3,\ p_2=0.5,\ p_3=0.15,\ p_4=0.05$

Then $P(X_1=2,\ X_2=4,\ X_3=1,\ X_4=1)=\dfrac{8!}{2!\times 4!\times 1!\times 1!}(0.3)^2(0.5)^4(0.15)^1(0.05)^1=0.0354$

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EXERCISE – 4.4

MULTINOMIAL DISTRIBUTION

1. Use the multinomial formula to find the probabilities also find $E(X_1)$ *and* $V(X_1)$.

$(a)\ n=5,\ X_1=1,\ X_2=2,\ X_3=2,\ p_1=0.3,\ p_2=0.6,\ p_3=0.1$

$(b)\ n=6,\ X_1=3,\ X_2=2,\ X_3=1,\ p_1=0.5,\ p_2=0.3,\ p_3=0.2$

$(Ans: (a)\ 0.0324,\ 1.5,\ 1.05 \qquad (b)\ 0.135,\ 3,\ 1.5)$

2. In a factory producing certain items, 30 per cent of the items produced have no defect, 40 per cent have one defect, and 30 per cent have two defects. A random sample of 8 items is taken from a day's output. Find the probability that it will contain 2 items with no defect, 3 items with one defect, and 3 items with two defects. $(Ans: 0.0871)$

3. The probabilities are 0.50, 0.40, and 0.10 that a trailer truck will have no violations, 1 violation, or 2 or more violations when it is given a safety inspection by state police. If 5 trailer trucks are inspected, find the probability that 3 will have no violations, 1 will have 1 violation, and 1 will have 2 or more violation. $(Ans: 0.1)$

4. According to a genetics theory, a certain cross of guinea pigs will result in red, black, and white offspring in the ratio 8:4:4. Find the probability that among 8 offspring, 5 will be red, 2 black, and 1 white. $(Ans: 21/256)$

5. The complexity of arrivals and departures of planes at an airport is such that computer simulation is often used to model the "ideal" conditions. For a certain airport with three runways, it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

$Runway\ 1: p_1=2/9, \qquad Runway\ 2: p_2=1/6, \qquad Runway\ 3: p_3=11/18$

What is the probability that 6 randomly arriving airplanes are distributed in the following fashion?

$Runway\ 1:\ 2\ airplanes, \qquad Runway\ 2:\ 1\ airplane, \qquad Runway\ 3:\ 3\ airplanes$

$(Ans: 0.1127)$

6. The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, automobile, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by automobile, and 2 arrived by train? $(Ans: 0.0077)$

7. According to Mendel's theory, if tall and colorful plants are crossed with short and colorless plants, the corresponding probabilities are $9/16, 3/16, 3/16$ *and* $1/16$ for tall and colorful, tall and colorless, short and colorful, and short and colorless, respectively. If 8 plants are selected, find the probability that 1 will be tall and colorful, 3 will be tall and colorless, 3 will be short and colorful, and 1 will be short and colorless. $(Ans: 0.002)$

8. When people were asked if they felt that the laws covering the sale of firearms should be more strict, less strict, or kept as they are now, 54% responded more strict, 11% responded less, 34% said keep them as they are now, and 1% had no opinion. If 10 randomly selected people are asked the same question, what is the probability that 4 will respond more strict, 3 less, 2 keep them the same, and 1 have no opinion? $(Ans: 0.0016)$

9. Find the probability of obtaining 2 ones, 1 two, 1 three, 2 fours, 3 fives and 1 six in 10 rolls of a balanced die. $(Ans: 0.0025)$

10. According to the manufacturer, M&M's are produced and distributed in the following proportions: 13% brown, 13% red, 14% yellow, 16% green, 20% orange, and 24% blue. In a random sample of 12 M&M's, what is the probability of having 2 of each color? $(Ans: 0.0025)$

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